

Intelligent Complaint Classification Using Localized Transformer Architectures

Background

As organizations increasingly rely on digital channels for customer support, they receive thousands of complaints through emails, chat platforms, web forms, and social media. These complaints contain valuable insights into customer experiences but are typically written as unstructured text. Manual review and routing require significant human effort, increase operational costs, and often delay issue resolution. These challenges become even greater in multilingual and culturally diverse environments where users employ informal language, slang, abbreviations, spelling variations, and code-switching.

Problem Statement

Customer support organizations struggle to efficiently classify large volumes of unstructured consumer complaints. Existing keyword-based or traditional machine learning approaches often fail to capture contextual meaning, emotional cues, and localized language patterns, leading to incorrect routing, inconsistent prioritization, and slower response times. This problem is particularly relevant in countries such as Nigeria, where complaints frequently contain informal English, Nigerian Pidgin, local expressions, abbreviations, and spelling variations that are poorly represented in existing NLP systems.

Research Gap

Transformer-based language models such as BERT have significantly improved text classification performance compared to traditional machine learning methods. However, most existing research focuses on standard English datasets or multilingual datasets from well-resourced languages. Limited work has explored complaint classification in African contexts where informal language, code-switching, and local linguistic expressions are common. Furthermore, many existing systems prioritize overall predictive performance without explicitly considering fairness and robustness across different language varieties and writing styles.

Research Aim

This research aims to develop a transformer-based complaint classification system capable of accurately categorizing consumer complaints containing informal language and localized linguistic expressions. The study will investigate whether domain-specific preprocessing and localized fine-tuning can improve classification performance compared to conventional NLP approaches.

Research Objectives

- Analyze consumer complaint datasets to identify common linguistic patterns and challenges.
- Design preprocessing techniques to handle noisy, informal, and localized text.
- Fine-tune transformer architectures such as BERT for complaint classification.
- Compare transformer-based models with traditional machine learning baselines.
- Evaluate model performance using Precision, Recall, Macro F1-score, and Accuracy.
- Assess model robustness on complaints containing localized language and informal expressions.

Scope

This research focuses exclusively on text-based consumer complaint classification.

The study excludes:

- Voice or speech-based complaints
- Image or multimedia complaints
- Complaint generation
- Automated response generation

Expected Contributions

The proposed system is expected to:

- Reduce manual complaint triage time.
- Improve complaint routing accuracy.
- Enable faster and more consistent customer support.
- Demonstrate the effectiveness of localized transformer architectures.
- Contribute insights toward NLP research for low-resource and linguistically diverse communities.

Ethical Considerations

Guided by the values of **responsibility** and **fairness**, this research will consider the ethical implications of deploying AI in customer support systems.

Key considerations include:

- Protecting customer privacy through data anonymization and responsible data handling.
- Ensuring fair model performance across different writing styles, dialects, and linguistic backgrounds.

- Maintaining human oversight for high-priority or sensitive complaints.
- Providing transparency regarding model confidence, limitations, and potential sources of error.

Research Questions

1. Can transformer-based architectures outperform traditional NLP methods for consumer complaint classification?
2. How does localized preprocessing influence classification performance?
3. Can localized transformer models better understand informal and code-mixed complaint text?
4. Which transformer architecture provides the best balance between predictive performance and computational efficiency?

Evaluation Metrics

The proposed system will be evaluated using:

1. Macro F1-score
2. Precision
3. Recall
4. Accuracy
5. Inference Time

Note: *This ordering reflects: classification quality under imbalance first, diagnostic error-type breakdown second, overall correctness third, and computational efficiency last (relevant for real-world/edge deployment of your localised model).*

References

1. Jain, P., Tripathi, S., Garg, T. et al. An intelligent transformer based framework for bilingual financial complaint classification. *Sci Rep* **16**, 20594 (2026).
<https://doi.org/10.1038/s41598-026-51771-w>